Ex.No: 7

24.9.26

Sliding Window Protocol

Aim:

To write sliding window protocol code using python.

Code:

Sender.Py

```
import time
import os
ws = input("Window Size")
msg = input("Enter messag:")
frame = [(i+1,ch) for i,ch in enumerate
                              (msg)]
i=0
while i < len(frames):
    window = frames[i:i+ws]
    with open('Sender-Buffer","w") as f:
        for no,data in window:
            f.write(f"{no},{data}\n")
    print("\n Sending :", window)
    time.sleep(2)
    while not os.path.exists("Receiver-
                              Buffer"):
        time.sleep(1)
    with open("Receiver-Buffer") as f:
        ack,no = f.read().strip().split(1)
        no = int(no)
    print("Received", ack, no)
    if ack == "ACK" and no == window[-1][0]
        i += ws
    elif ack == "NAck":
        i = no-1
    Open("Receiver-Buffer","w").close()
```

```
print("\n Transmission Completed")
```

Receiver.Py

```
import time.os
expected = 1
while true:
    if not os.path.exists("Sender-Buffer"):
        time.sleep(1)
        continue
    With open ("Sender-Buffer") as f:
        frames = [(int (x split(",")[0]),
            x.split(",")[1].Strip(1) for x in
                        f if x.strip()]
        if not frames:
            time sleep(1)
            continue
        print ("\n Recived ", frames)
        for (no, data) in frames:
            if no != expected:
                With open ("Receiver-Buffer", "w") as f:
                    f.write (f"NACK, {expected}")
                break
            expected += 1
        else:
            With open ("Receiver-Buffer", "w") as f:
                f.write(f"ACK, {frames[-1][0]}")
    time.Sleep(1)
```

IDLE Shell 3.11.9

File Edit Shell Debug Options Window Help

```
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:\Users\keshv\OneDrive\Desktop\cn\sender.py ==========
Enter Window Size: 4
Enter Text Message: Hello World

Frames sent:
Frame No: 1, DATA: H
Frame No: 2, DATA: e
Frame No: 3, DATA: l
Frame No: 4, DATA: l

Run receiver.py and press ENTER after Receiver_Buffer is created...
Receiver: ACK:1
Receiver: ACK:2
Receiver: ACK:3
Receiver: ACK:4

All frames acknowledged.

Frames sent:
Frame No: 5, DATA: o
Frame No: 6, DATA:
Frame No: 7, DATA: W
Frame No: 8, DATA: o

Run receiver.py and press ENTER after Receiver_Buffer is created...
Receiver: ACK:5
Receiver: ACK:6
Receiver: ACK:7
Receiver: ACK:8

All frames acknowledged.

Frames sent:
Frame No: 9, DATA: r
Frame No: 10, DATA: l
Frame No: 11, DATA: d

Run receiver.py and press ENTER after Receiver_Buffer is created...
Receiver: ACK:9
Receiver: ACK:10
Receiver: ACK:11

All frames acknowledged.

Transmission completed.
```

IDLE Shell 3.14.2

File Edit Shell Debug Options Window Help

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:\Users\keshv\OneDrive\Desktop\cn\receiver.py ==========
Received Frame: 1, DATA: H
Frame 1 accepted
Received Frame: 2, DATA: e
Frame 2 accepted
Received Frame: 3, DATA: l
Frame 3 accepted
Received Frame: 4, DATA: l
Frame 4 accepted

Receiver_Buffer updated.
>>>
========== RESTART: C:\Users\keshv\OneDrive\Desktop\cn\receiver.py ==========
Received Frame: 5, DATA: o
Frame 5 accepted
Received Frame: 6, DATA:
Frame 6 accepted
Received Frame: 7, DATA: W
Frame 7 accepted
Received Frame: 8, DATA: o
Frame 8 accepted

Receiver_Buffer updated.
>>>
========== RESTART: C:\Users\keshv\OneDrive\Desktop\cn\receiver.py ==========
Received Frame: 9, DATA: r
Frame 9 accepted
Received Frame: 10, DATA: l
Frame 10 accepted
Received Frame: 11, DATA: d
Frame 11 accepted

Receiver_Buffer updated.
```

Result:

Thus, the sliding window protocol was implemented successfully.